

SuperMoTo: an integrated monitoring matrix, room correction and Ambisonics decoding plugin for audio production and immersive listening rooms

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Abstract

SuperMoTo is an open-source audio plugin (VST3, AU and Standalone) that puts a studio’s monitoring signal chain into one plugin instance. From that instance it routes up to 32 inputs to 32 outputs through a configurable gain-and-EQ matrix, applies per-loudspeaker FIR correction filters derived from in-room acoustic measurements, time-aligns the loudspeakers of a rig and phase-aligns a subwoofer to the corrected main speakers, and generates periphonic Ambisonics decoders up to third order for immersive listening rooms. Acoustic measurement tools and a transfer-function analysis and correction design engine are built in, so the whole calibration workflow happens inside the plugin, one loudspeaker at a time or for a whole rig in a single pass. Measured on a stereo rig with one subwoofer at ten listening positions, through eight filter variants played side by side, the subwoofer alignment held the level through the crossover to within 0.03 dB when its design delay was shortened by 14 ms, where a minimum-phase correction lost 2.2 dB to 2.3 dB, and confining the phase correction to the crossover region removed 14 dB to 24 dB of pre-echo from 500 Hz to 8 kHz without changing the summation. It is implemented in C++ using the JUCE framework and is licensed under the AGPL 3.0, or under commercial terms for holders of a commercial JUCE licence.

1 Introduction

Decisions about balance, timbre and spatial imagery are all made at the listening position, so a systematic error in what the engineer hears accumulates through the whole mix. In a real room, what the engineer hears is never the loudspeaker alone but the combined response of the loudspeaker, the room, and any bass-management or routing in the signal path. Three independent classes of error contribute to this. Room modes below the Schroeder frequency produce large, position-dependent peaks and dips with audible ringing, speaker-boundary interference and early reflections add comb-filtering through the midrange, and the loudspeaker’s own magnitude and phase response is rarely flat [15]. Electronic correction cannot cure all of these (room cancellation notches and late reflections are not minimum-phase and cannot be inverted by any causal filter [4]), but the broad, reproducible, minimum-phase portion of the combined loudspeaker-room response can be usefully attenuated with a well-designed FIR filter, which reduces coloration and improves translation to other playback systems.

No single open-source tool covers this workflow end to end. The widely-used Room EQ Wizard (REW)¹ provides acoustic measurement and filter design as a standalone desktop application but not as a DAW plugin, requiring the resulting correction impulse responses to be loaded into a separate convolution plugin and routing to be managed outside the tool. Commercial room correction products such as Sonarworks SoundID Reference² and IK Multimedia ARC³ focus exclusively on correction and do not integrate routing, bass management, or immersive audio decoding. High-channel routing tools and Ambisonics de-

¹<https://www.roomeqwizard.com/>

²<https://www.sonarworks.com/soundid-reference>

³<https://www.ikmultimedia.com/products/arcstudio/>

coders exist as separate plugins, the IEM Plug-in Suite⁴ being the reference open-source example, but are not combined with a calibration workflow. SuperMoTo covers the whole workflow in one plugin instance, namely the monitoring matrix, the measurement engine, the correction design, the FIR convolution and the Ambisonics decoder generation. It can also read the decoding matrices exported by IEM’s AllRADe-coder, so an all-round decoder can be driven by a set of measured, individually corrected loudspeakers. The combination matters most in VR and immersive audio production, where a loudspeaker array has to be routed, time-aligned, individually corrected and driven by a decoded Ambisonics signal at the same time.

Aligning a subwoofer’s phase to the mains is itself established practice rather than a new idea, and Section 7 places the construction used here against the parametric all-pass designs it descends from. The contributions of this paper are the following.

- A monitoring chain that carries routing, per-loudspeaker EQ, time alignment, FIR correction and Ambisonics decoding in one plugin instance, with a latency budget that keeps a rig aligned exactly as the user set it for the least added latency (Section 3).
- A closed-form subwoofer phase alignment, read directly off the measured phase and carried by the correction filter that is already there, in place of a parametric all-pass fitted by an optimiser, as done by other solutions (Section 7). Measured, it makes the main-to-subwoofer summation independent of the delay it is given (Section 9).
- A power-weighted crossover-band group-delay estimator for the main-to-subwoofer offset, and the identity that makes one such offset per loudspeaker correct across a whole rig sharing a single subwoofer (Sections 7 and 8).
- A linear-phase correction whose phase is kept only through the crossover region, where the alignment needs it, and is minimum-phase above; measured, this removes the pre-echo a full phase inversion puts ahead of transients and costs the summation nothing (Sections 6 and 9).
- An all-in-one open-source implementation in which the measurement, the analysis, the filter design and the playback chain are the same instance, so a rig can be calibrated without leaving it (Section 10).

2 The monitoring chain

The routing matrix connects up to 32 inputs to 32 outputs through a grid of crosspoint *frames*. Six independent configurations (A–F) are stored per preset and can be engaged in exclusive mode (instant A/B switching between monitoring rigs) or summed. Everything the plugin does to the audio happens in one pass over the block, in the order the signal physically travels. With X_i the inputs and $\omega = 2\pi f$, the signal reaching output o is

$$Y_o(\omega) = g \underbrace{e^{-j\omega k_o/f_s}}_{\text{latency comp.}} \underbrace{\tau_o E_o(\omega) T(\omega) e^{-j\omega \delta_o/f_s} C_o(\omega)}_{\text{loudspeaker } o} \sum_{c \in \mathcal{A}} \sum_i \underbrace{\gamma_{cio} B_{cio}(\omega)}_{\text{crosspoint}} X_i(\omega), \quad (1)$$

where $\mathcal{A}$ is the set of engaged configurations, k_o and δ_o are delays in samples at the sampling rate f_s , $T(\omega)$ is an optional monitor target curve shared unmodified by every output, and g is the master gain (level, mute and dim on one smoother).

A crosspoint contributes a gain and polarity γ_{cio} , ramped linearly over 50 ms so that switching a route, or a whole configuration, cannot click, followed by its own two-band EQ B_{cio} , which is where a configuration-specific tweak on that one route belongs. The per-output terms are the trim and polarity τ_o , a second two-band EQ E_o , the target curve T , a delay of δ_o samples (up to 100 ms) for physical speaker time-alignment, and the FIR correction C_o , applied by a zero-latency partitioned convolution engine [16]. Each EQ band is a peaking biquad, or a lowpass, highpass or bandpass of second or fourth order, a fourth-order section being two biquads carrying the Butterworth Q pair scaled by the band’s own Q setting.

The FIR correction C_o is designed to flatten the measured response, which is a target in itself: a loudspeaker corrected flat at the listening position sounds harsh, because its directivity index rises with frequency [21] and the flattened measurement has therefore lifted the direct sound in the treble by the

⁴<https://plugins.iem.at/>, GPLv3, from the Institute of Electronic Music and Acoustics, Graz.

amount the room’s reverberant field would otherwise have rolled off. T is where that is put right: a gentle downward tilt from 0 dB at 20 Hz to a user-chosen fall (typically -3 dB to -6 dB) at 20 kHz, picked from factory curves informed by EBU Tech 3276’s operational room response mask [20], ITU-R BS.1116-3 [21] and the Harman in-room preference target [23, 22], or defined by the user as a straight tilt, an EBU-like turnover, or a low-shelf bass lift. Because T carries no output index o , it is identical on the mains and the subwoofer, so it cancels out of the phase relationship the alignment of Section 7 depends on, where a tilt folded into each output’s own correction filter C_o instead would not.

The delay is rounded to a whole number of samples and read from a ring buffer at one tap, so that it is a pure sample shift and the same operation on every output. The half sample given up is 11 μ s at 44.1 kHz, 4 mm of path, and one sample of relative error between two outputs places its first cancellation notch at $f_s/2$, so it cannot comb inside the band, where interpolating between two taps would colour the treble by where the delay falls between samples.

Everything that describes a physical loudspeaker therefore multiplies once, after the sum, and is shared by every configuration (A...F) that drives it, while the routing and its per-route tweaks belong to one configuration. When filters of different lengths are loaded on different outputs, automatic inter-output latency compensation keeps every speaker time-aligned at the listening position for the least added latency (Section 3).

A configuration tool generates standard stereo and surround layouts (2.0 through 7.1) with bass management and writes a complete matrix configuration ready for refinement. The subwoofer is fed either from a dedicated input (an LFE channel) or from the sum of the main channels, and bass management is written as a highpass on each main output and a lowpass on the subwoofer’s.

3 Latency and time alignment

The partitioned convolution engine adds no block latency whatever the filter length, but a linear-phase correction is centred on its own impulse, so output o carries a bulk latency ℓ_o equal to the position of the peak of its loaded impulse response (about half the filter length, and near zero for a minimum-phase filter). Filters of different lengths on different outputs would therefore introduce timing inconsistencies between them. Writing d_o for the delay the user (or automatic analysis) set on output o in samples and $(\cdot)_+$ for the positive part, the delay line actually runs only the remainder $\delta_o = (d_o - \ell_o)_+$, the uniform compensation is

$$k = \max_{o \text{ fed}} (\ell_o - d_o)_+, \quad (2)$$

and each fed output takes the share $k_o = d_o + k - \max(d_o, \ell_o)$ of it, so that

$$\delta_o + \ell_o + k_o = d_o + k \quad \text{on every fed output.} \quad (3)$$

The rig is aligned exactly as wanted, offset by a single common term k , which is the smallest value satisfying $d_o + k \geq \ell_o$ everywhere and is zero whenever each output’s own alignment delay already covers its own filter, the usual case for a rig that has been time-aligned. An output that no engaged configuration feeds is left out of the maximum, so an unused loudspeaker’s long filter cannot impose latency on the rig.

4 Ambisonics decoding

The configuration tool also generates periphonic (3D) Ambisonics sampling decoders for orders 1 through 3, following the AmbiX convention (ACN channel order, SN3D normalisation) [18]. For each loudspeaker at azimuth θ , elevation φ and radius r , the decode gain is

$$D[s][j] \propto a_{n(j)} Y_j^{\text{SN3D}}(\Omega_s), \quad (4)$$

where a_n are the $\max\text{-}r_E$ per-degree weights derived from the largest root of the Legendre polynomial P_{N+1} [19], which improves high-frequency localisation stability. The rig is described by a loudspeaker count and a table of azimuth, elevation, radius and output channel, starting either from a hand-tuned regular layout for the order’s canonical count or from a near-uniform spherical Fibonacci lattice for any other count. Non-equidistant speaker rigs are compensated by per-speaker delay and trim derived from the inverse-square law, each of which can be left out so that an alignment or a level already obtained

from measurements survives a later re-application of the decoder. Bass management low-passes the omnidirectional W component to a subwoofer and high-passes the speaker feeds.

Irregular and hemispherical rigs are better served by an all-round decoder than by a plain sampling decoder, so the tool can instead import the `.json` configuration exported by the IEM AllRADecoder and use its matrix, converting it to the plugin's SN3D/ACN feed and applying the same radius compensation.

5 Acoustic measurement

The measurement view drives the plugin outputs with a band-limited stimulus (5s to 30s duration) and records the microphone response on one of the plugin's input channels. The sweep stimulus is a *synchronized* exponential sweep in the sense of [2], its sweep rate being quantised so that the stimulus phase at each harmonic arrival time is an exact multiple of 2π , which is what allows the distortion orders to be separated with their true phase at analysis time (Section 6). White noise is offered as the alternative stimulus.

Each captured stereo file stores the raw stimulus on channel 1 and the microphone on channel 2, so the transfer function estimate at analysis time is always the ratio of the captured microphone signal to the sent stimulus. Three modes are provided, "Dry" (bare loudspeaker and room, correction bypassed), "FIR" (single output with its correction in circuit, for before and after verification), and "System" (complete engine including routing and all FIR filters, for end-to-end validation).

A run measures every selected channel in turn with the microphone in one place and writes its files into a measurement folder that numbers positions per channel automatically and tags an optional subwoofer channel. The folder carries a generated XML manifest recording, for every run, its time, its stimulus parameters, a free-text comment, the microphone calibration in effect (name and a verbatim copy of the file) and the dB SPL calibration if one has been made by reading a real sound level meter. The folder is therefore self-describing, and the group-alignment mode of Section 8 loads one in a single step. Microphone frequency-response calibration files in the REW/miniDSP plain-text format are supported and are applied at analysis time rather than baked into the recordings, so the calibration can be updated, or replaced by the copy embedded in a folder measured years earlier, without re-measuring.

6 Transfer-function estimation and correction design

For each set of measurements (one speaker, multiple microphone positions), the plugin offers two estimators for the loudspeaker-room transfer function. The first is the Welch H_1 estimator [1]

$$\hat{H}(f) = \frac{\hat{P}_{xy}(f)}{\hat{P}_{xx}(f)}, \quad (5)$$

with Hann-windowed 50%-overlapped segments and a user-selectable window length of 16384 to 262144 samples, and it applies to any stimulus. The second, available and selected by default when the run's manifest identifies it as a synchronized sweep, deconvolves the recording by the analytic spectral inverse of the sweep [2], which yields the full-band response from a single sweep and requires no regularised division by a measured spectrum. Because a distortion product of order m is emitted $L \ln m$ seconds earlier than the linear response at the same frequency (L being the sweep rate), the deconvolved response separates the orders along the time axis. The linear impulse response is windowed out for the estimate, and orders 2 to 5 are retained as distortion curves, power-averaged across positions. Under the Welch estimator those same products are instead folded into the linear estimate. The two methods share the entire downstream chain, so a set can be re-analysed either way for comparison.

Before averaging across positions, the bulk propagation delay of each measurement is estimated from the impulse response peak and removed, so position-dependent phase ramps do not cancel in the spatial average. The spatial average follows the practice of correcting the broad trend common to the listening area rather than a single-seat interference pattern [15, 3]. Its magnitude is the power mean over the positions, and its phase the argument of their complex mean, blended towards the minimum phase of that magnitude as the positions stop agreeing: the weight is the debiased vector coherence $(NR^2 - 1)/(N - 1)$, clipped to $[0, 1]$, of the N positions, R being the modulus of their complex mean over their RMS modulus.

The correction filter is the regularised complex inverse of the frequency-dependent-smoothed average. A variable fractional-octave smoothing is applied, where the octave fraction interpolates log-linearly from

an independent low-frequency setting to a high-frequency setting, allowing the correction to track room modes finely at low frequencies while responding only to broad trends above a few kilohertz [5]. Boosts are limited by a smooth tanh soft-knee ceiling to avoid ringing and headroom loss at room cancellation notches [6], and the correction fades back to unity outside a user-set analysis range, where neither the loudspeaker nor the measurement carries usable information.

For subwoofer-equipped systems, a second set of measurements (the sub at the same microphone positions) is loaded, and the phase of the main's correction is steered onto the subwoofer's across the crossover region by an all-pass factor, so that the two sum coherently there. That design is developed in Section 7.

The final correction spectrum is exported as a mono 32-bit float impulse-response wav file in either *linear-phase* (energy centred at $N/2$, enabling phase correction and subwoofer alignment at the cost of $N/2$ samples of latency) or *minimum-phase* (real-cepstrum method, magnitude-only correction with essentially zero bulk latency, suited to tracking sessions) form [17].

By default the linear-phase export keeps its designed phase only through the crossover region. With $\hat{c} = C/|C|$ and $\hat{m}$ the unit phasor of the minimum-phase response of magnitude $|C|$, it exports

$$C_{\text{lin}}(f) = |C(f)| \frac{w_p(f) \hat{c}(f) + (1 - w_p(f)) \hat{m}(f)}{|w_p(f) \hat{c}(f) + (1 - w_p(f)) \hat{m}(f)|}, \quad (6)$$

where $w_p = 1$ up to $2f_x$, falls as a raised cosine in $\log_2 f$ to 0 at $4f_x$ and stays there, f_x being the crossover. The magnitude is untouched, and from $4f_x$ up the export is the minimum-phase one. Only the part of the average's phase that every seat shares can be inverted everywhere [4, 3]; a linear-phase filter puts the part a seat does not share *ahead* of the direct sound, as a pre-echo. Above the crossover region nothing needs that phase, since the subwoofer alignment of Section 7 is confined to it. The limit was added after an unlimited export, heard on impact sounds, produced a short pre-reverberation that its minimum-phase twin did not, and Section 9 measures both effects: what the limit removes and what it leaves unchanged.

7 Subwoofer phase alignment

A main loudspeaker and a subwoofer only sum to a flat response through the crossover when they are in phase there [7]. A fourth-order Linkwitz-Riley lowpass contributes a group delay of $2\sqrt{2}/\omega_x$ at DC and at its corner, 6.4 ms at $f_x = 70$ Hz, and the low-frequency acoustic centre of a loudspeaker moves with frequency as well [14]. Several milliseconds of the offset are therefore built into the system, and no repositioning of the box removes them.

Steering a main's phase onto the subwoofer's through the crossover has been proposed by a few authors or constructors. Audyssey's phase equalisation cascades second-order all-pass sections onto the satellite channel and computes their pole angles and radii recursively, so as to minimise $\varphi_{\text{sub}} - \varphi_{\text{sat}} - \varphi_{\text{AP}}$ over a weighted band around the crossover [9, 10], and Dirac Live Bass Control applies a pre-specified number of all-pass biquads to the high- and low-frequency branches so that the subwoofer and the channel are in phase around the crossover at a chosen position in the room [11]. The crossover region remains an active subject [12]. In each of these the all-pass is a parametric filter of chosen order whose coefficients are found by an optimiser against a phase-error criterion. What follows shares that objective and differs in construction. The alignment is read straight off the measured phase in closed form, on the same frequency grid the correction is designed on, so there is no filter order to choose, no objective function and no convergence to check, and it is carried by the FIR that already applies the magnitude correction, so it costs nothing at run time.

Aligning the two first requires their relative timing to survive the analysis, which by default it does not. The bulk propagation delay of each measurement is estimated from its impulse-response peak and removed before the positions are averaged, since otherwise the position-dependent phase ramps cancel in the average. Each subwoofer capture is instead delay-anchored on the main capture taken at the same position, shifted by the main's delay rather than by its own, so the averaged subwoofer response $\bar{S}$ keeps its true phase relative to the main's average.

The alignment itself is a single factor applied to the correction spectrum. From the subwoofer's unit-magnitude phasor $\hat{s} = \hat{S}/|\hat{S}|$, where $\hat{S}(f) = \pm \bar{S}(f) e^{j2\pi fT}$ carries the subwoofer polarity and an assumed

bulk delay T discussed below, the correction C designed for the main is multiplied by

$$A(f) = \frac{(1 - w_x(f)) + w_x(f) \hat{s}(f)}{|(1 - w_x(f)) + w_x(f) \hat{s}(f)|}, \quad (7)$$

where the alignment weight is a raised cosine in log frequency,

$$w_x(f) = \frac{1}{2} + \frac{1}{2} \cos(\pi x(f)), \quad x(f) = \min\left(1, \max\left(0, \frac{1}{N_{\text{oct}}} \log_2 \frac{f}{f_x}\right)\right), \quad (8)$$

so $w_x = 1$ at and below the crossover f_x and $w_x = 0$ at and above $2^{N_{\text{oct}}} f_x$, with a release width N_{oct} of one octave by default. Above the release the main dominates and keeps the phase its own correction gives it.

Three properties of (7) are what make it usable. Taking the phasor discards the subwoofer's magnitude, so none of the subwoofer's own response leaks into the main's correction. The renormalisation makes $|A| = 1$ at every bin, so the alignment is strictly a phase operation and leaves the designed magnitude correction untouched, which is what allows the same design to serve with and without a subwoofer. And the blend is taken on the phasors rather than on an unwrapped phase angle, so the interpolation travels the short arc between 1 and $\hat{s}$ and the rotation A applies is bounded by π at every frequency, by construction and with no phase-unwrapping step, which would need an anchor in a band where the subwoofer is often below the noise. Scaling an unwrapped angle, anchored at the crossover, turns out to be equivalent once the bulk delay below has been removed, and worse when it has not.

The delay T is what keeps the all-pass small. Advancing the subwoofer phasor by T before its phase is taken leaves the all-pass carrying only the residual, while the bulk of the offset is applied as an ordinary delay on the main's output, where it costs nothing. T is a relative quantity, not the delay applied to the main. If the outputs receive bulk delays a_m and a_s , the phase difference left between the two acoustic paths through the crossover region is $2\pi f [T - (a_m - a_s)]$, which vanishes at every frequency only if

$$T = a_m - a_s. \quad (9)$$

The group-alignment mode designs each speaker's correction with exactly its own $a_i - a_s$, so (9) holds for every speaker of the rig and for any trim applied to the subwoofer against the group.

T is estimated from the measurements themselves, and the estimator has to be chosen with care, because the two available measures of lateness are not measuring the same quantity. An arrival time read from an impulse-response peak is accurate to about the inverse of the measured bandwidth [13], some 50 μs for a full-range loudspeaker but 10 ms for a subwoofer that passes nothing above 100 Hz, so it is weakest exactly where it is wanted. The plugin instead fits the slope of the anchored subwoofer's unwrapped phase over $[f_x/2, 2f_x]$ by least squares weighted by the measured power $|\hat{S}|^2$, which returns the effective group delay of the subwoofer in the crossover region, geometry and filtering together, and that is what decides whether the two sources add or cancel. Weighting each point of that fit by $|\hat{S}|^2$ lets the part of the band where the subwoofer actually has output decide the slope, which is what makes the estimate stable.

Because A carries phase, the correction is a mixed-phase filter and is not causal on its own, which is why the exported impulse response is centred at $N/2$ and why it brings the bulk latency the compensation of Section 3 then absorbs. A is the identity from $2f_x$ up, so the phase limit (6), which starts at $2f_x$, leaves it whole. A minimum-phase export rebuilds the filter from $|C|$ alone and A has unit magnitude, so the subwoofer alignment disappears from it entirely. The loudspeakers still arrive together in that mode, since the delays are written to the outputs regardless, but there is no phase steering through the crossover.

8 Multi-speaker group alignment

For rigs of more than a couple of loudspeakers, a group-analysis mode holds one transfer-function estimate per speaker (up to 16, plus one shared subwoofer when the rig has one) side by side, sharing a single set of correction-design settings across all of them. A whole measurement folder is loaded in one step, the manifest supplying which channels were measured, which one is the subwoofer and which output each speaker sits on. Because every channel of a run was measured with the microphone in one place, each

subwoofer capture is paired with the speaker capture of the same run, which keeps the pairing correct even when a speaker missed a run. Individual runs can be excluded from the analysis, and runs measured through an existing correction are excluded by default, since a correction has to be designed from the uncorrected loudspeaker.

Each speaker’s bulk propagation delay, estimated as in Section 6, is compared across the group and every driver is delayed to match the most distant one, so the whole rig arrives coherently at the reference position. The resulting delay is fed back into each speaker’s own subwoofer phase-alignment term before its correction filter is finalised. A level-matching trim per speaker is also derived, as the mean power of its corrected response over the 500 Hz to 2000 Hz band that SMPTE ST 2095-1 specifies for channel level calibration [8], relative to the quietest speaker of the group. It is reported rather than applied, so that a deliberate level offset is never overwritten.

The subwoofer itself is never given a correction FIR filter. It contributes only its computed alignment delay and polarity (plus eventual gain lowpass order 2/4 crossover filter as given by the config tool or tuned manually in the matrix).

Applying the result exports every speaker’s correction impulse response, writes the delays, the filters and the subwoofer’s polarity onto the corresponding outputs, and saves a markdown report listing the settings, the files used, the measured and applied delays, the suggested trims and, optionally, the plotted responses. Loading, re-analysing and exporting a whole group run on a background thread pool with progress feedback, so the editor stays responsive with long, multi-position measurement sets across many speakers, and a batch already running survives the editor being closed.

9 Validation

Rig and campaign. The rig is a two-way stereo pair with one subwoofer in a treated studio. The loudspeakers and the reference position form a triangle with 1.70 m sides and a 2.55 m base, and the subwoofer stands on the floor between the loudspeakers, 1.64 m from the reference position. Each main and the subwoofer were first measured on their own, correction bypassed, at ten microphone positions over the listening area (six spanning 45 cm along the axis and 34 cm across it, four of them repeated 6 cm higher) and once more at one of them with a chair taken out of the room, with one 20 s synchronized sweep per channel. Two group analyses were run on these eleven captures, one crossed over at 70 Hz with the analysis range starting at 40 Hz and one at 100 Hz from 60 Hz, both smoothed from 1/6 octave in the bass to 1/3 in the treble under a 12 dB boost ceiling. They were exported as eight variants. At 70 Hz, with the design delay the estimator of Section 7 returned, $T = 28.8$ ms: the linear-phase export at 8192 taps with and without the phase limit (6), the limited one at 4096 taps, and the minimum-phase export. At 100 Hz: the limited linear-phase and the minimum-phase exports at 8192 taps, each with the estimated $T = 32.2$ ms and with $T = 17.9$ ms, set by hand 14.3 ms shorter because it made the all-pass rotation smallest on the plugin’s phase plots.

All eight were loaded side by side in one matrix, each on three outputs of its own fed from two inputs of its own, and the whole system was measured at the ten positions, the sixteen inputs in turn at each position with 10 s sweeps. Any two variants are therefore compared at the same placement of the microphone, minutes apart. The subwoofer’s routing level, fitted on these runs, is -2.6 dB at 70 Hz and -0.25 dB at 100 Hz, and its polarity is normal in every variant, the inverted fit being 2 dB RMS worse. One variant, the limited 8192-tap one at 70 Hz, was measured with a main’s correction filter left active by mistake on its subwoofer output from an earlier session. The captures showed it before the saved plugin state did: the difference between that variant and the unlimited export, whose mains are the same filter below $2f_x$, is identical on both mains (correlation 0.999 over 30 Hz to 120 Hz) and worth -0.8 dB through the crossover band. With the stray filter, read back from the saved state, in the subwoofer’s term, (10) predicts -0.85 dB. The variant is used as measured above 250 Hz, where the subwoofer plays no part. Below, the unlimited export stands in for it.

By default, the plugin processes the entire sweep at once to get each response. Using Welch averaging on the same sweeps instead creates a wavy pattern (ripple) in the volume. This ripple repeats every 0.81 octaves (which matches the segment hop time) and gets worse as the system’s delay increases. In the impulse response it reads as echoes on both sides of the direct sound, some 20 dB above the real ones in the 250 Hz to 1000 Hz octaves, in every variant alike.

The model. Each dry subwoofer capture is delay-anchored on the main capture of the same position, so the same propagation delay d_p has been removed from both, and the acoustic sum at position p is exactly

$$Y_p(\omega) = e^{-j\omega T} H_{\text{hp}}(\omega) C(\omega) H_p(\omega) \pm g H_{\text{lp}}(\omega) S_p(\omega) \quad (10)$$

up to the common factor $e^{-j\omega d_p}$, which changes no magnitude. H_p and S_p are the measured responses of the main and of the subwoofer, H_{hp} and H_{lp} the bass-management filters, C the exported correction and g the subwoofer's routing level. Nothing else is modelled. Against all sixteen measured cases (eight variants, two mains, the unwanted stray filter included) the prediction is within 0.84 dB to 1.00 dB RMS over 35 Hz to 250 Hz (Figure 1), although the dry captures were taken two days before the system runs, and with positioning error of the order of the centimeter. A difference between two variants, which is what the comparisons below are made of, it predicts better still. An offline re-implementation of the analysis reproduces the plugin's sixteen exported filters to 0.02 dB RMS in magnitude from 40 Hz to 16 kHz. It is used where the campaign could not measure, *i.e.* between and beyond the measured delays, with the subwoofer inverted, and with a position left out of the design.

Latency and treble. The four variants at 70 Hz carry filters whose bulk latency is 4096, 2048 and 0 samples. At every position their mains arrived at the same sample, give or take two samples of peak picking (0.05 ms). The 100 Hz variants arrived 3.40 ms later and 10.88 ms earlier than those, exactly the differences between their output delays once each is rounded to a whole sample. The phase type leaves the treble alone: from 200 Hz to 10 kHz the linear- and minimum-phase exports of one design gave the same response to within 0.02 dB to 0.05 dB RMS, on both mains and at both crossovers. An earlier campaign on this rig, which included an uncorrected run, measured the minimum-phase correction as applied to within 0.31 dB and 0.33 dB RMS of the designed one over the same band.

Summation through the crossover. Table 1 compares variants two by two through the crossover band, at the same microphone placements, and sets each difference against the one (10) predicts from the same exported filters. At the estimated delays the all-pass carries 0.2 dB more level through the band than the minimum-phase export at 70 Hz, at all ten positions, and 0.4 dB to 0.5 dB more at 100 Hz, at eight. Given the hand-set delay, 14.3 ms shorter, the all-pass result moves by 0.01 dB to 0.03 dB while the minimum-phase one loses 2.2 dB to 2.3 dB at every position, a cancellation that Figure 1(c) shows at 110 Hz. Each of these differences is predicted to within 0.11 dB.

variants	difference	measured		predicted		positions
		left	right	left	right	
70 Hz, $T = 28.8$ ms	all-pass – minimum phase	+0.24	+0.18	+0.26	+0.22	10, 10
100 Hz, $T = 32.2$ ms	all-pass – minimum phase	+0.41	+0.48	+0.49	+0.59	8, 8
100 Hz, $T = 17.9$ ms	all-pass – minimum phase	+2.72	+2.64	+2.81	+2.61	10, 10
100 Hz, all-pass	$T = 32.2 - T = 17.9$ ms	+0.03	+0.01	+0.03	+0.04	
100 Hz, minimum phase	$T = 32.2 - T = 17.9$ ms	+2.34	+2.16	+2.35	+2.07	10, 10
70 Hz, all-pass	4096 – 8192 taps	−0.01	+0.01	−0.03	−0.02	

Table 1: Level through the crossover band $[f_x/\sqrt{2}, f_x\sqrt{2}]$, relative to 500 Hz to 2000 Hz: difference between two variants measured at the same microphone placements, mean over the ten positions, in dB, against the difference (10) predicts from the same exported filters. The last column counts the positions at which the first variant is the louder, where the difference is not nil. At 70 Hz the all-pass is the unlimited 8192-tap export, the same filter as the limited one below $2f_x$.

The delay. Figure 2 runs the design delay from 10 ms to 40 ms through the offline chain, with the measured variants on top. The all-pass stays between −0.24 dB and −0.29 dB of summation efficiency at 100 Hz and between −0.34 dB and −0.97 dB at 70 Hz. Over the same range the minimum-phase design swings with a period of about $1/f_x$, from −0.22 dB to −8.4 dB and from −0.40 dB to −6.4 dB. The all-pass advantage is not necessarily a better optimum (indeed, at its best delay the minimum-phase design sums as well). Instead, it eliminates the need to determine that delay. The estimator of Section 7 comes within about a millisecond of it here, which already costs the minimum-phase design 0.4 dB to 0.7 dB. The ripple left at 70 Hz, 0.6 dB every 10 ms, comes from the release band of (8). Near 100 Hz, $w_x \approx \frac{1}{2}$. When the residual phase at this frequency approaches 180°, the numerator of (7) nearly vanishes, causing

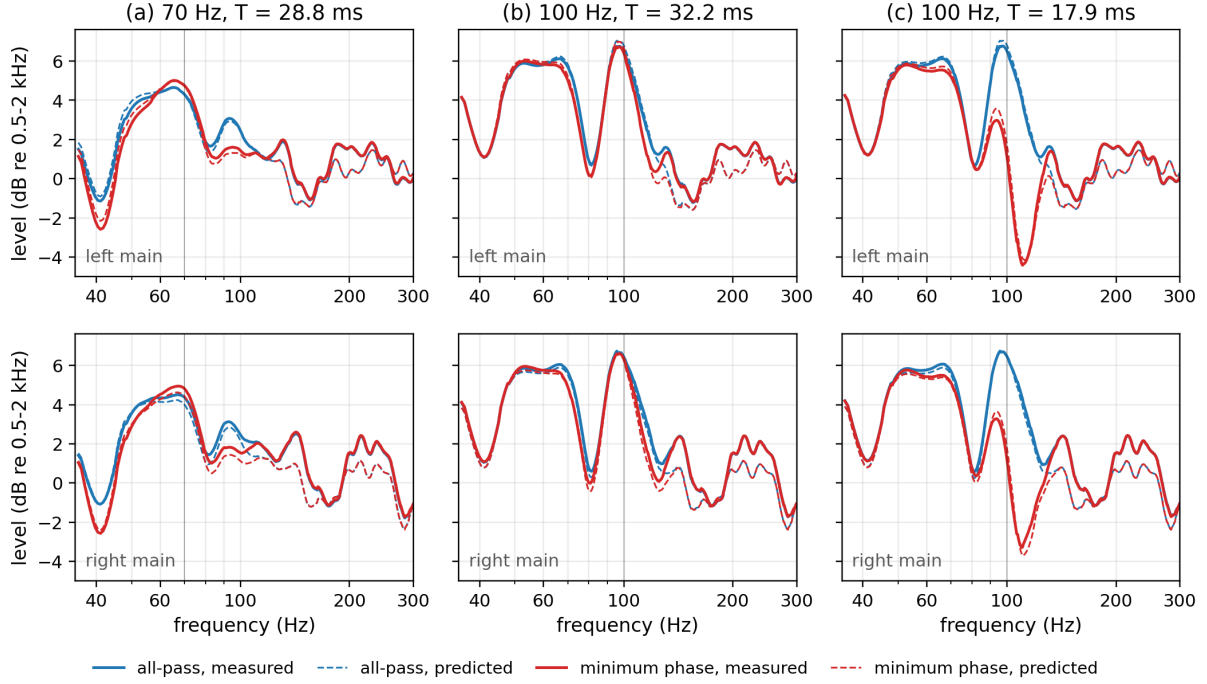


Figure 1: Level through the crossover region, relative to 500 Hz to 2000 Hz and averaged over the ten positions, left main (top) and right main (bottom), for the aligned linear-phase export (all-pass) and the minimum-phase one: measured (solid) and predicted by (10) from the dry captures and the exported filters (dashed). (a) 70 Hz, estimated delay; (b) 100 Hz, estimated delay; (c) 100 Hz, delay set by hand. The vertical line marks the crossover.

a sudden phase jump. Even with a 70 Hz crossover, the subwoofer still contributes significant energy at 100 Hz. Consequently, at $T = 26$ ms, the combined response drops by 3 dB in the 100 Hz to 110 Hz band, while remaining unaffected elsewhere. Inverting the subwoofer polarity across all three measured settings has minimal impact on the all-pass design, maintaining it between -0.26 dB and -0.50 dB. In contrast, this inversion degrades the minimum-phase design, dropping its score to between -1.9 dB and -6.0 dB. Finally, because the designs were optimized using nine positions and evaluated at a tenth, the fact that every out-of-sample score remains within 0.02 dB of its in-sample value proves a key point, which the fact that the alignment is a robust property of the entire listening area, and not an artifact of the specific measurement positions used to shape it.

What the delay still decides. The all-pass makes the steady-state sum independent of T , not the time response. T sets when the mains speak relative to the subwoofer, which no delay holds back, and therefore how much of the subwoofer’s energy arrives ahead of the mains’ direct sound. On top of that, the all-pass steers the main onto the subwoofer in linear phase, partly by advancing it. The lower row of Figure 2 shows both in the 125 Hz octave, measured and predicted. At 100 Hz, the 3 ms to 30 ms before the direct sound hold -4 dB of the energy that follows it with the estimated delay, and -13 dB to -16 dB with the hand-set one, against -15 dB and -31 dB for the minimum-phase export; at 70 Hz, -16 dB to -17 dB against -26 dB. The shorter delay has its own price: the subwoofer’s octave, 63 Hz, then peaks 17 ms after the direct sound at the median seat, against 3.5 ms with the estimated delay. The estimator aims at the delay a delay-only alignment needs. The all-pass does not need it, so T is free to be chosen for the time response instead, and whether early energy in the crossover band or late energy in the subwoofer’s band is the more audible is a listening question this campaign does not answer.

The phase limit. Table 2 measures what the limit (6) removes. From 500 Hz to 8 kHz the unlimited export put 14 dB to 24 dB more energy than the limited one in the 30 ms before the direct sound, and 14 dB to 30 dB more in the last 3 ms, in every octave on both mains. The limited export is within 2 dB of the minimum-phase one over the longer window and within 5 dB over the shorter, both approaching the noise floor above 1 kHz. At 250 Hz, inside the limit’s transition from $2f_x$ to $4f_x$, it removes 3 dB to 10 dB

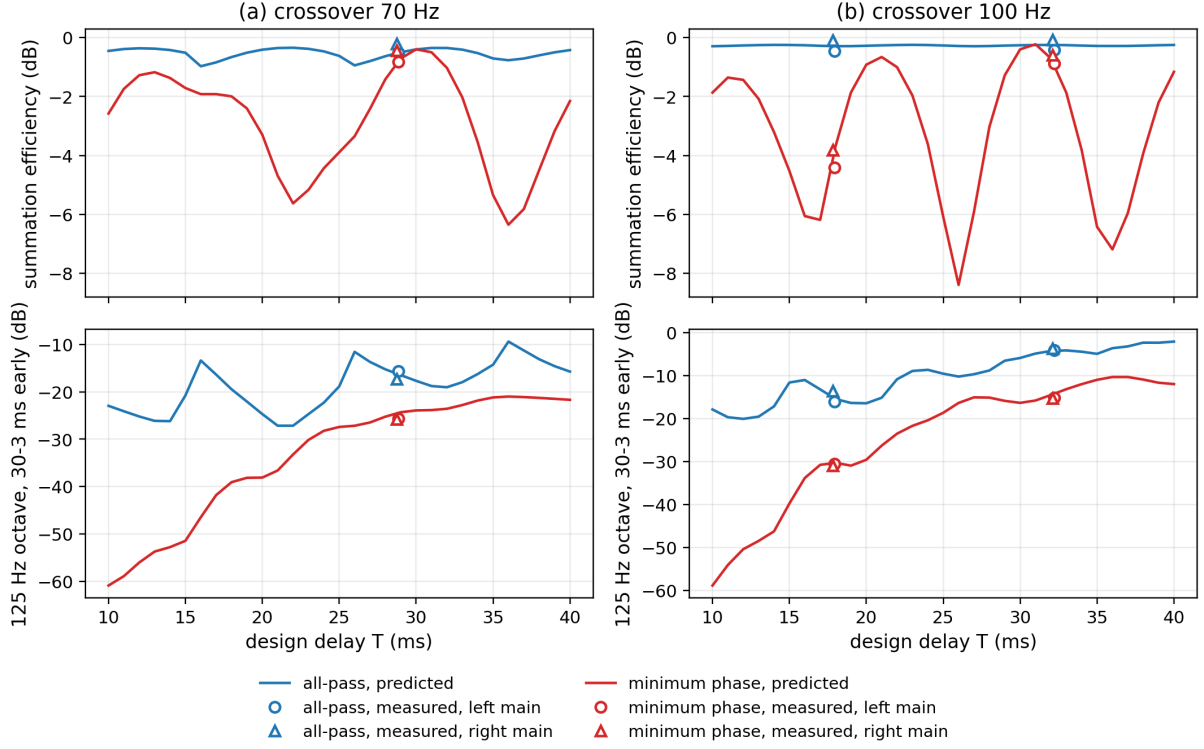


Figure 2: The design delay T swept through the offline chain (lines, mean of both mains), with the measured variants (markers, one per main). Top: summation efficiency over $[f_x/2, 2f_x]$, $20 \log_{10} |M + S| - 20 \log_{10}(|M| + |S|)$ weighted by the overlap of the two sources, 0 dB being coherent addition; a measured point takes $|M + S|$ from the system run and $|M| + |S|$ from the dry captures. Bottom: energy in the 125 Hz octave arriving 3 ms to 30 ms before the direct sound, relative to the octave’s energy after it.

and leaves 14 dB to 16 dB more than minimum phase. The inversion it removes bought no timing. From 500 Hz to 4 kHz every octave’s envelope peaked within 0.2 ms of the direct sound at the median position, in every variant and in the dry captures alike, and within 0.3 ms at the worst position in every variant. The unlimited export moved no octave from 250 Hz up by more than 0.11 ms against the minimum-phase one. Furthermore, this limit does not negatively impact the summation. Indeed, below $2f_x$ the limited and unlimited exports are the same filter, and the limited 4096-tap export sums within 0.01 dB of the unlimited 8192-tap one through the crossover band (Table 1).

Below $2f_x$ the linear-phase filter still puts energy ahead of the direct sound, -14 dB in the 63 Hz octave and -16 dB to -17 dB in the 125 Hz one at 70 Hz, against -20 dB and -26 dB in minimum phase. That is the alignment of the previous paragraphs, and only there did the linear phase move an arrival: the 125 Hz octave by 0.4 ms to 0.5 ms earlier and the 63 Hz one by 0.9 ms later.

Filter length. At 70 Hz, halving the filter to 4096 taps changed the shape of the response through the crossover band but not its level. The shorter filter left a narrow dip at 81 Hz, -1.5 dB and -1.8 dB on the two mains, with rises of 0.2 dB to 0.6 dB on either side, so that the level averaged over the band moved by 0.01 dB (Table 1). Equation (10), given each filter as rendered, predicted that difference curve to 0.04 dB RMS over 35 Hz to 250 Hz.

Limits. This is one rig in one room. The comparisons between the all-pass and the minimum-phase exports, between the two delays at 100 Hz and between the phase limit on and off are measured. The delay sweep between and beyond the measured points, the inverted subwoofer and the leave-one-out scores are predictions of (10), whose accuracy on this rig Table 1 and the measured points of Figure 2 establish. No listening test was run: what is measured is energy and time, and the audibility of what remains below $2f_x$ is left open.

	250 Hz	500 Hz	1 kHz	2 kHz	4 kHz	8 kHz
<i>30 ms to 3 ms before the direct sound</i>						
minimum phase	-49/-50	-57/-57	-64/-66	-68/-67	-68/-68	-67/-67
linear, limited	-35/-34	-56/-56	-64/-64	-67/-67	-68/-68	-67/-67
linear, not limited	-32/-24	-38/-34	-45/-42	-46/-47	-44/-48	-50/-53
noise floor	-68	-71	-72	-71	-68	-67
<i>3 ms to 1 ms before</i>						
minimum phase	-62/-65	-56/-59	-64/-63	-59/-56	-74/-69	-77/-61
linear, limited	-42/-44	-56/-58	-64/-63	-59/-56	-69/-69	-73/-61
linear, not limited	-39/-26	-28/-28	-36/-33	-40/-42	-39/-39	-43/-47

Table 2: Energy arriving ahead of the direct sound, per octave, relative to that octave’s energy in the 300 ms after it, in dB, through causal octave filters so that nothing is moved earlier than it arrived; power mean over the ten positions, left/right main, crossover 70 Hz, 8192 taps. The noise floor is the same 27 ms window taken 400 ms to 150 ms ahead of the direct sound, beyond the reach of any filter here.

The script that produces the tables and figures of this section from the campaign’s folder is in the repository named in Section 10.

10 Implementation and availability

SuperMoTo is written in C++ and built with CMake against the JUCE framework⁵. Its reusable parts (the correction and measurement DSP, the plotting components, the preset system) live in a separate library, FxmeTools⁶, shared with the author’s other plugins, and the parts that need no JUCE at all are compiled and unit-tested on their own. Per-output FIR convolution uses the WDL non-uniformly partitioned convolution engine⁷ [16].

The engine holds no parameters of its own. It pulls them from the shared model whenever the model’s version counter changes, under a try-lock that defers the pull to the next block if the editor happens to be writing, so a GUI edit or a state restore can never stall the audio callback, and every update path is written to allocate nothing on the audio thread. Impulse-response files are read, resampled and installed on the message thread under a lock that the audio thread only ever takes with a try-lock, so a load in flight costs a few unconvolved blocks rather than a blocked callback. Meters and the spectrum analyser are fed by lock-free taps placed on each crosspoint and each output.

Continuous-integration builds for Linux, macOS (Apple Silicon universal binary) and Windows are provided by a GitHub Actions workflow triggered by version tags. The plugin is available in VST3, AU, and Standalone formats with a fixed 32-in/32-out discrete channel bus, which sidesteps the inconsistent per-host discrete-channel-count negotiation of JUCE’s AU and VST3 wrappers, the active matrix size being an independent, user-selectable setting up to that count. A user manual covering the workflow, the controls and the signal processing behind them is written in LaTeX alongside the source.

The source, the manual, the unit tests and the scripts that produce Tables 1 and 2 and Figures 1 and 2 are at <https://github.com/odoare/SuperMoTo>, under the AGPL 3.0 or commercial terms; the manual, the paper and the scripts themselves are under the LGPL 3.0.

11 Conclusion

SuperMoTo puts the parts of a monitoring chain that are usually separate tools into one plugin instance, so that the routing, the per-loudspeaker correction, the time alignment and the Ambisonics decoding of a listening room are described in one place and calibrated without leaving it. The subwoofer alignment is the part that gains most from that arrangement. Because the measurement, the spatial average and the filter design live in the same instance, the alignment can be read straight off the measured phase in closed form and folded into the correction filter, where a tool working from outside the signal chain would have to fit a separate parametric all-pass to it.

⁵<https://juce.com/>

⁶<https://github.com/odoare/FxmeTools>

⁷<https://www.cockos.com/wdl/>

What the approach cannot do is bounded by what any causal filter can do in a room. Cancellation notches and late reflections are not minimum-phase and are not inverted here, and the correction is designed for the area the microphone positions span rather than for one specific position. Within those bounds the measurements of Section 9 support two claims. The subwoofer alignment makes the summation through the crossover independent of the delay and of the polarity it is given, where a minimum-phase correction depends on both by several decibels. At its best delay the minimum-phase correction does as well, so what the alignment removes is the need to find that delay, not a limit on the optimum (at least on the tested rig). And a room correction's phase belongs in the crossover region. Inverting it above bought no timing on this rig and put 14 dB to 24 dB more energy ahead of the direct sound than the minimum phase, which the limit removes at no cost to the summation. The delay the alignment is designed with still decides the low-frequency time response, and choosing it for that is the question the campaign leaves open.

12 Acknowledgements

Generative AI (Claude, Anthropic) was used in the preparation of this work: to improve english in parts of the text, to improve prior art, and to help writing the analysis scripts of Section 9. All measurements were made by the author, who checked the analysis against the plugin's own implementation and against the measured system, verified the cited literature, and takes full responsibility for the content of the paper. Design, modelling, and DSP implementation of SuperMoTo was performed by the author. Threading, GUI, and project managment was worked on with the help of that same AI tool.

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